AMARDEEP MEHTA, PhD

AI SYSTEMS & DISTRIBUTED SYSTEMS RESEARCHER

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PROFESSIONAL SUMMARY

Computer systems researcher and ML systems engineer with 10+ years of experience designing, prototyping and optimizing distributed systems, cloud/edge platforms, AI infrastructure, model-serving systems and GPU/NPU-backed runtimes. Research and engineering work spans distributed computing, resource allocation, LLM inference, accelerator runtime systems, model-weight movement, asynchronous execution, profiling, memory management and collective communication. At Huawei, led systems research and C++/Python development for LLM serving and accelerator runtimes; at Ericsson Research, developed distributed runtime, placement and edge-cloud systems and led research collaborations. Currently leading technical architecture for shared AI/ML infrastructure across AWS EKS and on-prem GPU environments at Lenovo. PhD in Computing Science with publications and patents in distributed and cloud systems.

TECHNICAL SKILLS

Research Areas: Distributed systems, ML systems, AI inference, edge/cloud systems, resource allocation, runtime systems, collective communication

Programming & Systems: Python, C, C++ (primary); Rust, Go, Java; Linux, gRPC

AI/ML Systems: PyTorch, vLLM/vLLM-Ascend, LLM serving, model pipelining, GPU/NPU runtime integration, profiling, memory management

Cloud & Platforms: Kubernetes, AWS EKS, Docker, ClearML, OpenStack; hybrid cloud/on-prem infrastructure, workload orchestration

RESEARCH & PROFESSIONAL EXPERIENCE

ML Infrastructure Lead | Lenovo AI Technology Center **Jul 2026 – Present**

- Leading the technical architecture and phased roadmap for shared AI/ML infrastructure spanning AWS EKS and on-prem GPU environments, including heterogeneous GPU workload orchestration and model-serving requirements.
- Designing and evaluating Kubernetes/AWS EKS runtime patterns for heterogeneous GPU workloads, including resource allocation, priority queues, tenant isolation, data locality and capacity planning.
- Assessing model-serving architectures, GPU capacity and deployment patterns for internal AI workloads, including vLLM-based LLM inference.
- Leading technical evaluation and deployment planning of ClearML Enterprise as an ML control plane, covering storage, networking, access control and multi-cluster integration requirements.

Principal Engineer | Huawei Edinburgh Research Center **Sep 2022 – Jun 2026**

- Led systems research and hands-on C++/Python development for ML infrastructure, LLM serving and GPU/NPU accelerator runtime systems.
- Designed and implemented a C++ multipath model-transfer library integrated with vLLM-Ascend, using scheduling, chunking and asynchronous execution to improve LLM model-weight movement.
- Characterised AI/LLM workloads across latency, throughput, compute, communication, memory transfer and runtime bottlenecks to guide systems optimisation.
- Developed model-pipelining techniques to overlap model execution and transmission, reducing model loading and switching overhead for LLM workloads.
- Designed gRPC-based accelerator API remoting and scheduling mechanisms, batching asynchronous runtime operations to reduce communication overhead and improve GPU/NPU utilisation.
- Developed PyTorch extensions and plugins for low-overhead profiling and GPU/NPU memory management, improving visibility into performance bottlenecks and workload switching.
- Developed simulation and optimisation methods for collective communication across generic network topologies for distributed ML workloads.

Experienced Researcher | Ericsson Research — Cloud Systems & Platforms — Stockholm **Oct 2018 – Aug 2022**

- Designed and prototyped high-performance distributed systems and cloud/edge platforms using C++, Python, Rust and Erlang.
- Architected a scalable runtime and naming service for decentralised process orchestration across heterogeneous edge infrastructure.
- Developed resource-placement and availability optimisation methods for distributed cloud environments.
- Led an edge-cloud architecture study focused on cost-efficient placement of compute resources across distributed infrastructure.
- Led research collaborations, supervised Master's students and contributed to four patent filings in distributed and cloud systems.

EARLIER EXPERIENCE

Research Intern | Ericsson Research — Cloud Systems & Platforms — Lund / Stockholm **2015 – 2016**

- Ported and optimised an IoT software stack for resource-constrained devices using C and Python.
- Developed a Python-based framework for resource orchestration and auto-scaling in a hybrid cloud operating-system environment.

Software Developer | Ericsson — IP Multimedia Systems — Stockholm **Aug 2012 – Jul 2013**

- Developed C++ features and test infrastructure for the Multimedia Telephony Application Service, supporting system robustness and reliability.

EDUCATION

PhD, Computing Science | Umeå University — Sweden **2013 – 2018**

- Thesis: Resource Allocation for Mobile Edge Clouds.
- Developed scalable resource-allocation methods for applications running across heterogeneous distributed edge infrastructure using Python.
- Built a workload-analysis framework to characterise application resource requirements and support placement and capacity-planning decisions.

Micro-Degree, Data Science | KTH & Ericsson — Sweden **2020 – 2021**

- Coursework included applied machine learning, deep neural networks and machine learning in production.

MSc, Computer Science | Uppsala University — Sweden **2010 – 2012**

- Thesis: Methods for improving Voice Activity Detection in Communication Services.

BEng, Information Technology | Panjab University — Chandigarh, India **2006 – 2010**

- Major project: Audio Digital Watermarking (Java).

PATENTS & SELECTED PUBLICATIONS

Patents: Four filings in distributed/cloud systems, including two granted patents: “Methods and apparatuses for providing a function as a service” and “Synchronizing conflict-free replicated data types”.

- Mehta, A. & Eleftheriadis, L. (2022). Smart Edge Power Management to Improve Availability and Cost-efficiency of Edge Cloud. IEEE CLOUD.
- Nguyen, C., Mehta, A., Klein, C. & Elmroth, E. (2019). Why cloud applications are not ready for the edge (yet). ACM/IEEE SEC.
- Mehta, A. & Elmroth, E. (2018). Distributed cost-optimized placement for latency-critical applications in heterogeneous environments. IEEE ICAC.
- Mehta, A., Baddour, R., et al. (2017). Calvin constrained — a framework for IoT applications in heterogeneous environments. IEEE ICDCS.

Additional publications and research details available on request.